

Executive Summary: BulkMatrix is an intelligent bulk cargo procurement and vessel planning platform designed to model capesize/panamax shipping dynamics to East Coast Indian ports (Paradip, Dhamra, Vizag, Haldia, Gopalpur, Gangavaram). Following an in-depth codebase audit of package manifests, Express controllers, Python forecasting engines, and Docker configs, this report establishes a lean, high-performance, non-redundant technology stack strictly adhering to the "Do Not Blindly Add" governance mandate.

A Current Technology Stack (Codebase Verified)

LAYER	TECHNOLOGIES VERIFIED	CODEBASE EVIDENCE & FILE ANCHORS
Frontend	React 19, Vite 8, Tailwind CSS v4, Recharts, Axios, React Router v7, Leaflet, GSAP	frontend/package.json : react@19.2.8 , @tailwindcss/vite , recharts@3.10.1 , axios@1.20.0 , leaflet@1.9.4 , gsap@3.15.0
Backend	Node.js (ESM), Express.js 5, Cookie-Parser, CORS, Dotenv, Nodemailer	backend/package.json : express@5.2.1 , cookie-parser , cors ; backend/server.js mounts 14 REST routes
Database	MongoDB Atlas, Mongoose 9	backend/config/db.js connects to Mongo URI; 16 models in backend/models/ (Charter, Vessel, Port, Alert, WeatherRisk)
Authentication	JWT (jsonwebtoken 9), bcrypt 6	backend/middleware/authMiddleware.js : Bearer/cookie token extraction & RBAC (logistics_manager , vessel_owner , admin)
AI / ML Service	Python 3.10+, FastAPI, Pandas, NumPy, CatBoost, Scikit-learn	backend/main.py runs FastAPI on port 8000; src/prediction_pipeline.py & src/charter_engine.py execute pipeline
Model Storage	Python Pickle (.pkl binaries)	models/regularized/ contains 168 pre-trained per-route CatBoost models (catboost_reg_*.pkl) loaded via pickle.load()
Dev & DevOps	Docker, Docker Compose, Git & GitHub	docker-compose.yml orchestrates 4 services (mongodb, ml-service, backend, frontend); Dockerfile.ml containerizes Python

B Recommended Technology Stack

Frontend Tier

APPROVED

- **React 19 + Vite:** Instant HMR and optimal SPA bundle delivery.
- **Tailwind CSS v4:** Hardware-accelerated styling for dense data views.
- **Recharts & Leaflet:** Interactive BDI curve forecasting & voyage tracking.
- **Axios:** Pre-configured with automatic JWT bearer interceptors.

Backend API Gateway

APPROVED

- **Node.js + Express 5:** Non-blocking I/O API gateway.
- **REST Endpoints:** Clean REST APIs for charters, fleet, and analytics.
- **mlService.js:** Resilient proxy to Python ML service with mock fallback.
- **Nodemailer:** Transactional charter executed alerts.

Database & Persistence

APPROVED

- **MongoDB Atlas:** Managed multi-region document store.
- **Mongoose 9:** Strongly typed schemas for nested charter legs, vessel draft specifications, and port tidal restrictions.

AI / ML Forecasting Engine

APPROVED

- **Python 3.10+ & FastAPI:** Asynchronous microservice on port 8000.
- **Pandas & Scikit-learn:** Preprocessing, imputation, and scoring.
- **CatBoost Regressors:** 15, 30, 90-day horizon rate predictions.
- **Pickle Serialization:** Direct loading of 168 pre-trained model files.

Authentication & Security

APPROVED

- **JWT + bcrypt:** Cryptographically signed tokens with salted passwords.
- **RBAC:** Role enforcement across managers, owners, and administrators.

Deployment & DevOps

APPROVED

- **Vercel:** Static SPA edge distribution for React/Vite frontend.
- **Render:** Cloud hosting for Node.js backend and Python ML engine.
- **Docker:** Containerized local stack & reproducible build images.
- **GitHub:** Version control & CI/CD trigger for Vercel/Render.

D Technologies Worth Retaining & Solidifying

Docker & Docker Compose

RETAINED

Justification: BulkMatrix is a polyglot architecture (Node.js + Python ML + MongoDB). Docker isolates heavy Python machine learning dependencies (CatBoost, LightGBM, Statsmodels) and enables identical local development and cloud container execution.

Scikit-learn, Pandas & CatBoost

RETAINED

Justification: Actively executed inside `src/prediction_pipeline.py` for data normalization, route feature engineering, and evaluating multi-horizon freight rate predictions for Capesize/Panamax vessels.

Python Pickle (.pkl)

RETAINED

Justification: Directly used to deserialize the 168 pre-trained per-route model artifacts stored in `models/regularized/` with zero extra wrapping libraries.

GitHub Version Control & CI/CD

RETAINED

Justification: The authoritative source repository triggering automated previews on Vercel and zero-downtime container deployments on Render.

E Implementation: Concrete Gaps Resolved

Auditor Fix 1: ML Dependency Reconciliation (`requirement.txt`)

The ML service (`backend/main.py`) is powered by FastAPI, but `requirement.txt` was missing `fastapi`, `uvicorn`, and `pydantic`. Any container build with `Dockerfile.ml` failed. Added `fastapi ≥ 0.100.0`, `uvicorn ≥ 0.23.0`, and `pydantic ≥ 2.0.0` to ensure reliable container compilation.

Auditor Fix 2: Production Containerization (`frontend/nginx.conf`, `frontend/Dockerfile`)

Configured a dedicated production Nginx configuration with single-page application fallback (`try_files $uri $uri/ /index.html;`), eliminating 404 errors on deep SPA navigation. Updated `docker-compose.yml` to properly map host port 5173 to container port 80.

C Technologies to Remove / Not Add (Strict Technical Rationale)

TECHNOLOGY	AUDIT VERDICT	ARCHITECTURAL RATIONALE
Supabase	DO NOT ADD	Redundant. MongoDB + Mongoose is already deeply integrated with 16 active schemas. Introducing Supabase (PostgreSQL + GoTrue) forces dual databases, breaks referential integrity, and adds zero capability.
Firebase	DO NOT ADD	Redundant. Authentication is handled cleanly by JWT + bcrypt, and persistence is on MongoDB. Adding Firebase SDKs duplicates existing auth logic and locks the system into Google vendor constraints.
React Native	DO NOT ADD	Out of scope. BulkMatrix is an enterprise desktop procurement platform with wide multi-port route tables, Leaflet maps, and complex scenario planners. It is strictly a web application.
Java / Spring Boot	DO NOT ADD	Architectural bloat. The existing Express.js gateway and Python FastAPI service already satisfy all requirements. Adding Java introduces JVM memory overhead and fragments backend logic across 3 runtimes.
Next.js	DO NOT ADD	Unwarranted rewrite. BulkMatrix is an authenticated, internal procurement single-page application built on Vite. SSR/Server Components add caching complexity without providing SEO benefits behind auth.
Google Cloud (GCP)	DO NOT ADD	Over-engineering. Vercel (frontend) + Render (Node backend + Python ML container) + MongoDB Atlas provide automated CI/CD and cost predictability. Migrating to GCP introduces needless IAM and DevOps friction.
Framer Motion	DO NOT ADD	Bundle bloat. <code>frontend/package.json</code> already includes GSAP (<code>gsap@3.15.0</code>) and Tailwind CSS transitions for smooth UI animations. Adding Framer Motion adds 30KB+ of dead weight.
Redis	DO NOT ADD	Premature optimization. Python holds model objects in memory and MongoDB handles rate indexes. Concurrency in bulk procurement workflows does not warrant the cost and complexity of a dedicated Redis cluster.
Prisma	DO NOT ADD	Mismatched ORM. Mongoose is natively tailored for MongoDB schemas, discriminators, and population. Prisma is primarily designed for relational SQL and offers inferior MongoDB capability.
Botpress	DO NOT ADD	Mismatched function. The existing chat controller (<code>chatController.js</code>) handles human-to-human charter negotiations between cargo logistics managers and vessel owners. Botpress is a conversational AI bot and cannot replace contractual negotiation.
Joblib	DO NOT ADD	Unused. The 168 pre-trained models in <code>models/regularized/</code> are serialized strictly with Python's standard <code>pickle</code> library. Adding Joblib adds no value.

F Final BulkMatrix System Architecture



1. CLIENT TIER • React 19 + Vite + Tailwind CSS SPA (Deployed on Vercel)

Recharts BDI forecasting curves • Leaflet Indian East Coast maritime route maps • Axios with auto JWT Bearer interceptor

↓ HTTPS / REST API JSON (Authenticated via Bearer Token)



2. API GATEWAY • Node.js + Express 5 REST API (Deployed on Render)

JWT authentication & RBAC (Logistics Manager, Vessel Owner, Admin) • Controllers for Charters, Fleet, Contracts, Weather, Chat

↓ Mongoose ODM (mongodb:// cluster connection)



3. PERSISTENCE TIER • MongoDB Atlas (16 Mongoose Schemas)

Charters, Vessels, Ports (Paradip, Dhamra, Vizag, Haldia), WeatherRisk, PortCongestion, Contracts, ChatMessages

↓ Internal HTTP Microservice Proxy (backend/services/mlService.js → Port 8000)



4. ML INFERENCE ENGINE • Python FastAPI Microservice (Render Docker Service)

charter_engine.py & prediction_pipeline.py • Pandas feature extraction, draft/beam port compatibility constraints

↓ Deserialization via Python pickle.load()



5. SERIALIZED MODEL STORE • 168 Pre-Trained CatBoost Models (models/regularized/*.pkl)

15-day, 30-day, and 90-day per-route freight rate regressors • Multi-Vessel Fleet Optimization & Buy/Hold charter signals

End-to-End Operational Lifecycle:

- Voyage Parameter Entry:** Logistics Manager inputs cargo volume (e.g., 75,000 MT coking coal), origin port (e.g., Hay Point, Australia), and target East Coast destination ports (Paradip / Dhamra).
- Client Dispatch:** Axios dispatches an authenticated JSON payload to the Express API gateway with user JWT.
- Policy & Gatekeeping:** Express verifies JWT claims, validates cargo tonnage, and forwards the payload to the internal ML proxy (backend/services/mlService.js).
- Inference Execution:** FastAPI invokes `charter_engine.py`, extracts historical BDI lag indicators using Pandas, checks draft and beam compatibility against Indian port depth constraints, and executes the regularized CatBoost model.
- Output Delivery:** The system computes Buy/Hold market signals, port waiting time estimates, and Capesize vs. Panamax cost trade-offs, persisting the record in MongoDB and returning high-confidence analytics to the frontend.